

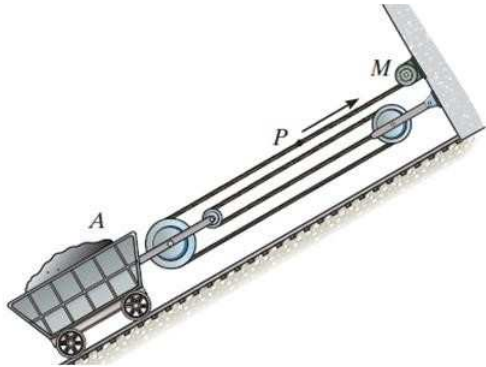
동역학

(Engineering Mechanics: Dynamics)

Chapter 12-4

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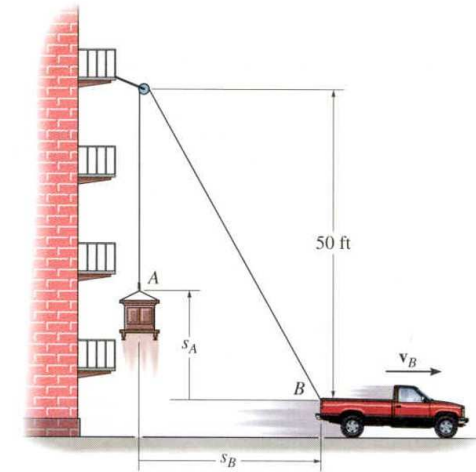
Absolute Dependent Motion Analysis of Two Particles



The cable and pulley system shown can be used to modify the speed of the mine car, A, relative to the speed of the motor, M.

It is important to establish the relationships between the various motions in order to determine the power requirements for the motor and the tension in the cable.

For instance, if the speed of the cable (P) is known because we know the motor characteristics, how can we determine the speed of the mine car? Will the slope of the track have any impact on the answer?



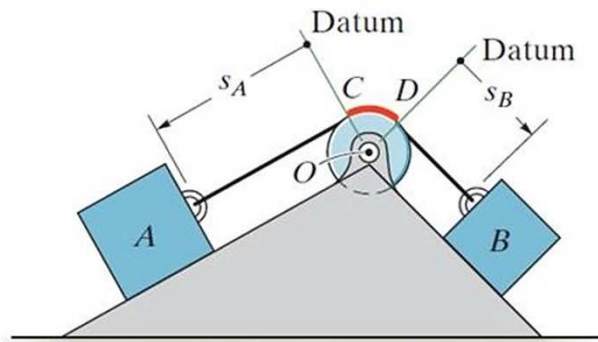
Rope and pulley arrangements are often used to assist in lifting heavy objects. The total lifting force required from the truck depends on both the weight and the acceleration of the cabinet.

How can we determine the acceleration and velocity of the cabinet if the acceleration of the truck is known?



Dependent Motion

In many kinematics problems, the motion of one object will depend on the motion of another object.

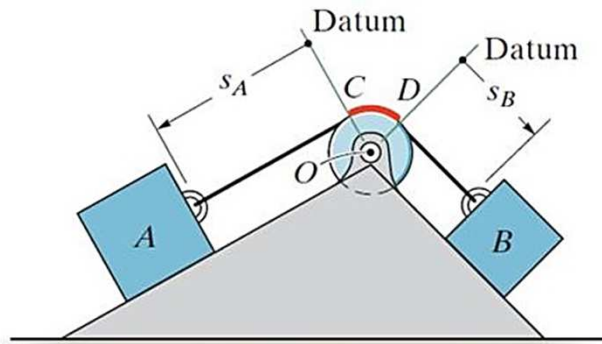


The blocks in this figure are connected by an inextensible cord wrapped around a pulley. If block A moves downward along the inclined plane, block B will move up the other incline.

The motion of each block can be related mathematically by defining position coordinates, s_A and s_B . Each coordinate axis is defined from a fixed point or datum line, measured positive along each plane in the direction of motion of each block.



Dependent Motion



In this example, position coordinates s_A and s_B can be defined from fixed datum lines extending from the center of the pulley along each incline to blocks A and B.

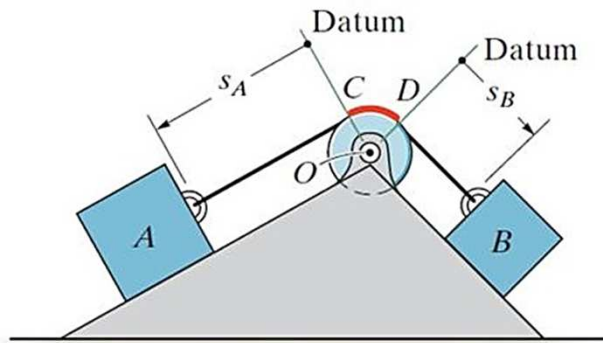
If the cord has a fixed length, the position coordinates s_A and s_B are related mathematically by the equation

$$s_A + l_{CD} + s_B = l_T$$

Here, l_T is the total cord length and l_{CD} is the length of cord passing over the arc CD on the pulley.



Dependent Motion



The velocities of blocks A and B can be related by differentiating the position equation. Note that l_{CD} and l_t remain constant, so

$$dl_{CD}/dt = dl_t/dt = 0$$

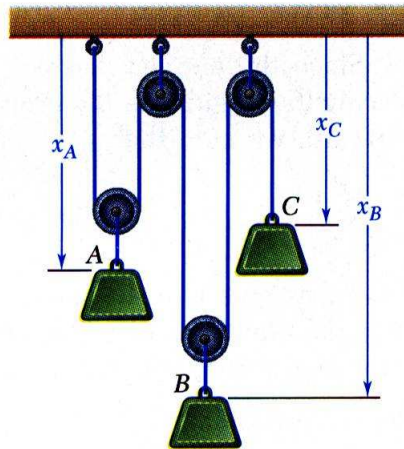
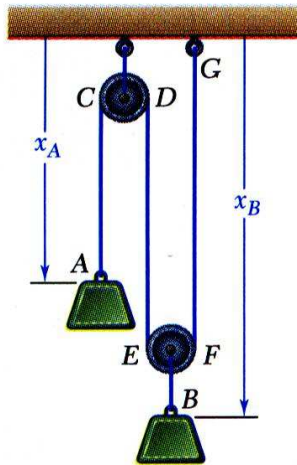
$$ds_A/dt + ds_B/dt = 0 \Rightarrow v_B = -v_A$$

The negative sign indicates that as A moves down the incline (positive s_A direction), B moves up the incline (negative s_B direction). Accelerations can be found by differentiating the velocity expression. Prove to yourself that

$$a_B = -a_A.$$



Dependent Motion Example I: Pulley System



- Position of a particle may *depend* on position of one or more other particles.
- Position of block B depends on position of block A . Since rope is of constant length, it follows that sum of lengths of segments must be constant.

$$x_A + 2x_B = \text{constant} \quad (\text{one degree of freedom})$$

- Positions of three blocks are dependent.

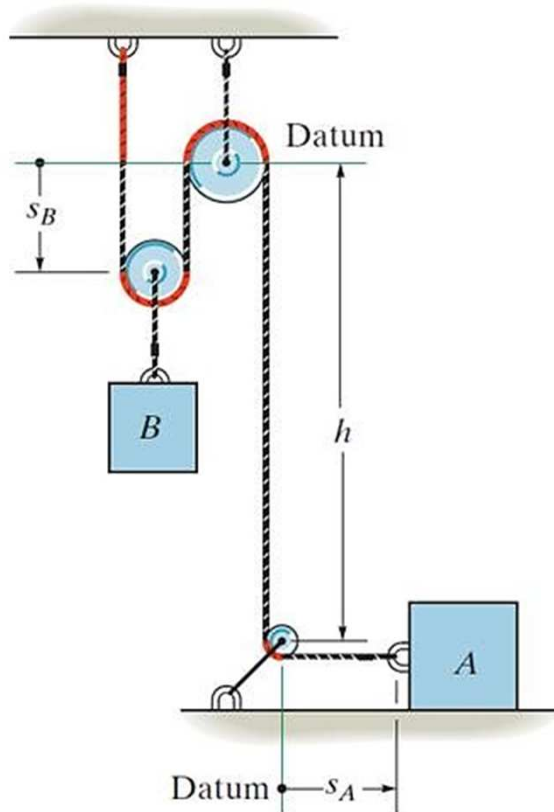
$$2x_A + 2x_B + x_C = \text{constant} \quad (\text{two degrees of freedom})$$

- For linearly related positions, similar relations hold between velocities and accelerations.

$$2 \frac{dx_A}{dt} + 2 \frac{dx_B}{dt} + \frac{dx_C}{dt} = 0 \quad \text{or} \quad 2v_A + 2v_B + v_C = 0$$

$$2 \frac{dv_A}{dt} + 2 \frac{dv_B}{dt} + \frac{dv_C}{dt} = 0 \quad \text{or} \quad 2a_A + 2a_B + a_C = 0$$

Dependent Motion Example II: Pulley System



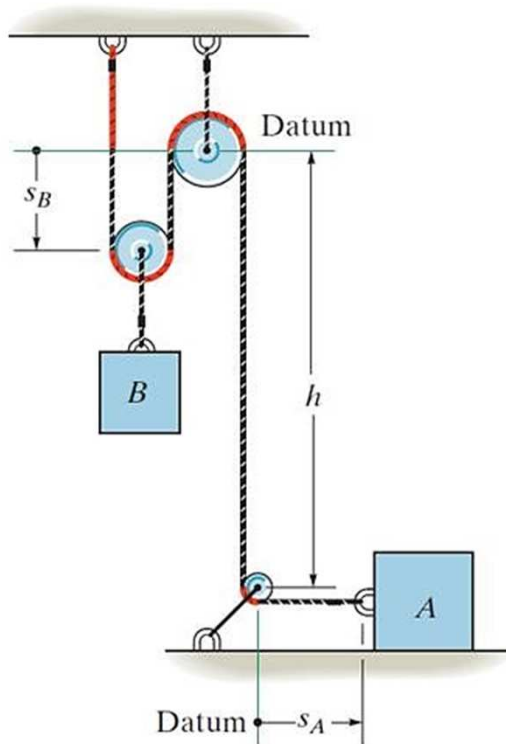
Consider a more complicated example. Position coordinates (s_A and s_B) are defined from fixed datum lines, measured along the direction of motion of each block.

Note that s_B is only defined to the center of the pulley above block B, since this block moves with the pulley. Also, h is a constant.

The red-colored segments of the cord remain constant in length during motion of the blocks.



Dependent Motion Example II: Pulley System



The position coordinates are related by the equation

$$2s_B + h + s_A = l_T$$

Where l_T is the total cord length minus the lengths of the red segments.

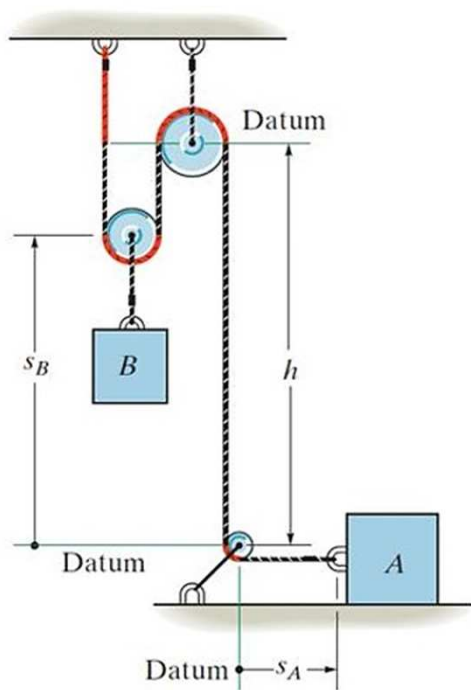
Since l_T and h remain constant during the motion, the velocities and accelerations can be related by two successive time derivatives:

$$2v_B = -v_A \quad \text{and} \quad 2a_B = -a_A$$

When block B moves downward, $(+s_B)$ block A moves to the left $(-s_A)$

Remember to be consistent with your sign convention!

Dependent Motion Example II: Pulley System



This example can also be worked by defining the position coordinate for B (s_B) from the bottom pulley instead of the top pulley.

The position, velocity, and acceleration relations then become

$$2(h - s_B) + h + s_A = l_T$$

$$\text{and } 2v_B = v_A \quad 2a_B = a_A$$

Prove to yourself that the results are the same, even if the sign conventions are different than the previous formulation.



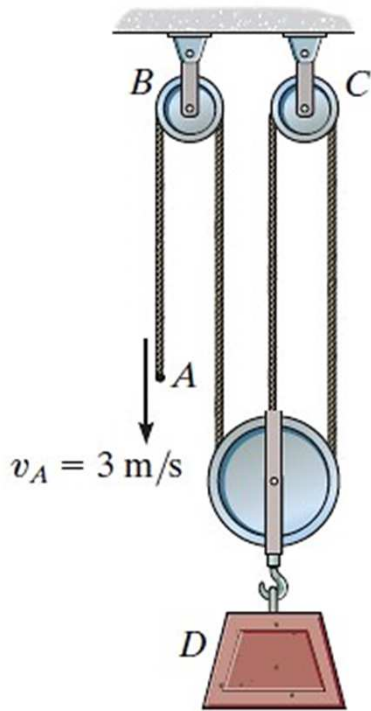
Dependent Motion: Procedures

These procedures can be used to relate the dependent motion of particles moving along rectilinear paths (only the magnitudes of velocity and acceleration change, not their line of direction).

1. Define position coordinates from fixed datum lines, along the path of each particle. Different datum lines can be used for each particle.
2. Relate the position coordinates to the cord length. Segments of cord that do not change in length during the motion may be left out.
3. If a system contains more than one cord, relate the position of a point on one cord to a point on another cord. Separate equations are written for each cord.
4. Differentiate the position coordinate equation(s) to relate velocities and accelerations. Keep track of signs!



Example: Block hanged by a Pulley



Given: In the figure on the left, the cord at A is pulled down with a speed of 3 m/s.

Find: The speed of block D.

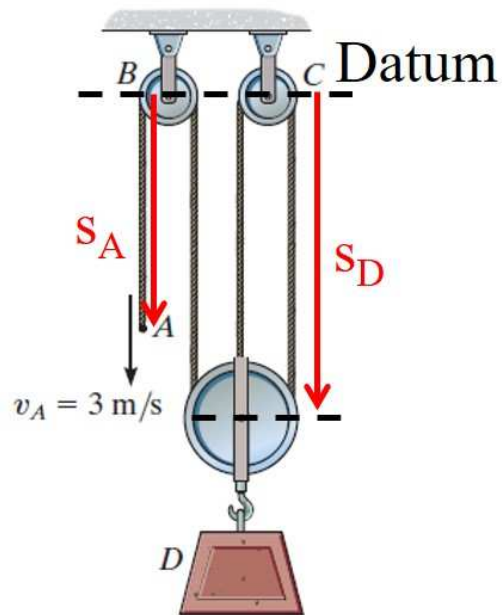
Plan: There is only one cord involved in the motion, so only one position/length equation is required. Define position coordinates for block D and cable lengths that change, write the position relation and then differentiate it to find the relationship between the two velocities.



Example: Block hanged by a Pulley

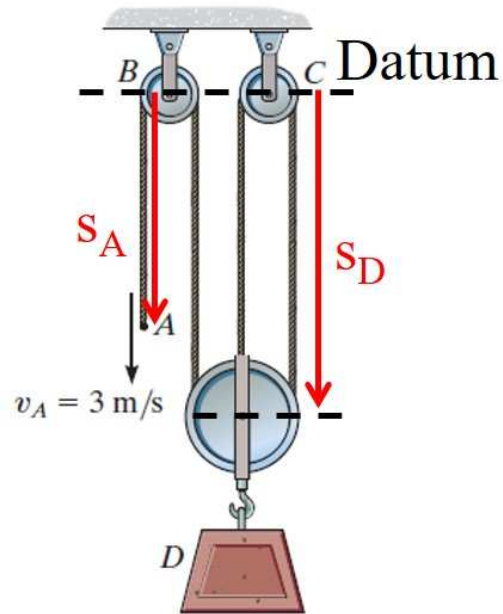
Solution:

- 1) A datum line can be drawn through the upper, fixed pulleys. Two coordinates must be defined: one for block D (s_D) and one for the changing cable length (s_A).



- s_A can be defined to the point A.
- s_D can be defined to the center of the pulley above D.
- All coordinates are defined as positive down and along the direction of motion of each point/object.

Example: Block hanged by a Pulley



- 2) Write position/length equations for the cord. Define l_T as the length of the cord, minus any segments of constant length.

$$s_A + 3s_D = l_T$$

- 3) Differentiate to find the velocity relationship:

$$v_A + 3v_D = 0$$

Since the cord at A is pulled down with a speed of 3 m/s,

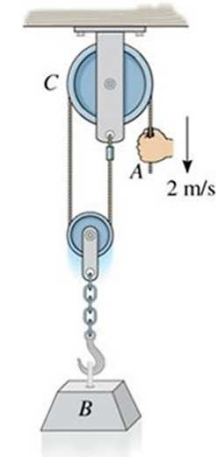
$$3 + 3v_D = 0 \Rightarrow v_D = -1 \text{ m/s} = 1 \text{ m/s} \uparrow$$



Concept Quiz

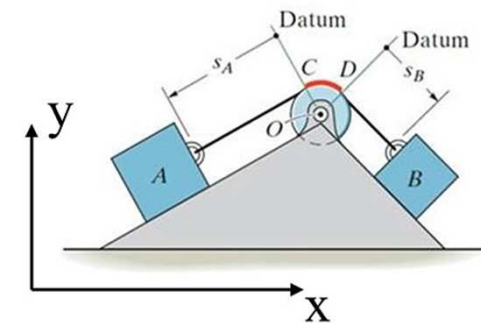
1. Determine the speed of block B.

- A. 1 m/s B. 2 m/s
C. 4 m/s D. None of the above.



2. Two blocks are interconnected by a cable. Which of the following is correct?

- A. $(v_x)_A = -(v_x)_B$ B. $v_A = -v_B$
C. $(v_y)_A = -(v_y)_B$ D. All of the above.

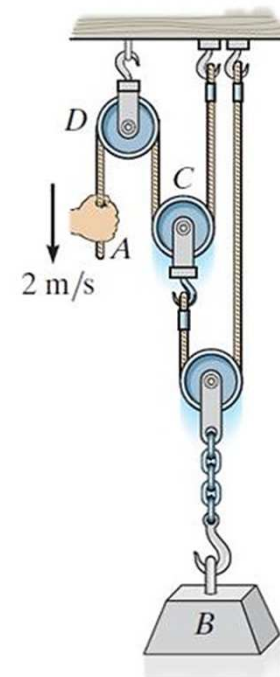


Group Problem Solving I: Pulley System

Given: In the figure on the left, the cord at A is pulled down with a speed of 2 meter per second.

Find: The speed of block B.

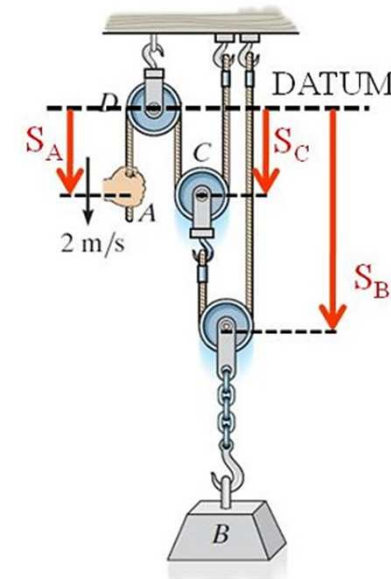
Plan: There are two cords involved in the motion in this example. There will be two position equations (one for each cord). Write these two equations, combine them, and then differentiate them.



Group Problem Solving I: Pulley System

Solutions:

- 1) Define the position coordinates from a fixed datum line. Three coordinates must be defined: one for point A (s_A), one for block B (s_B), and one for block C (s_C).
 - Define the datum line through the top pulley (which has a fixed position).
 - s_A can be defined to the point A.
 - s_B can be defined to the center of the pulley above B.
 - s_C is defined to the center of pulley C.
 - All coordinates are defined as positive down and along the direction of motion of each point/object.



Group Problem Solving I: Pulley System

- 2) Write position/length equations for each cord. Define l_1 as the length of the first cord, minus any segments of constant length. Define l_2 in a similar manner for the second cord:

$$\text{Cord 1 : } s_A + 2s_C = l_1$$

$$\text{Cord 2 : } s_B + (s_B - s_C) = l_2$$

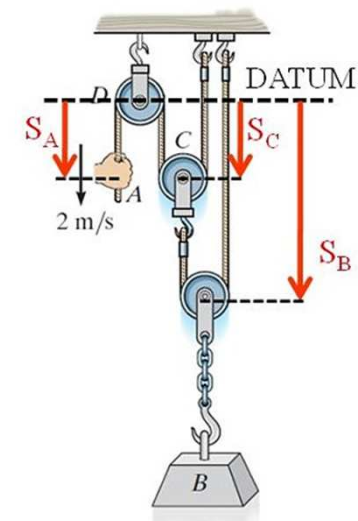
- 3) Eliminating s_C between the two equations, we get

$$s_A + 4s_B = l_1 + 2l_2$$

- 4) Relate velocities by differentiating this expression. Note that l_1 and l_2 are constant lengths.

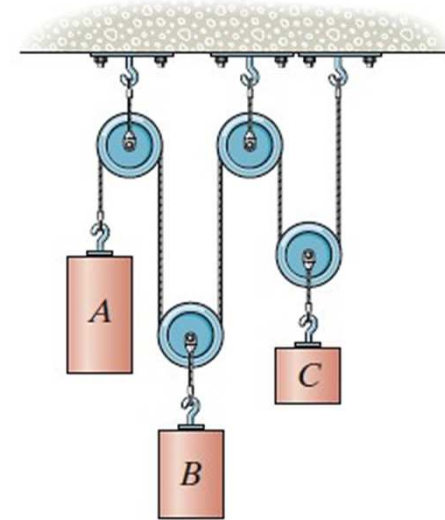
$$v_A + 4v_B = 0 \Rightarrow v_B = -0.25v_A = -0.25(2) = -0.5 \text{ m/s}$$

The velocity of block B is 0.5m/s up (negative s_B direction).



Group Problem Solving II: Pulley System with 3 Blocks

Given: In this pulley system, block A is moving downward with a speed of 6 meter per second while block C is moving down at 18 meter per second.



Find: The speed of block B.

Plan: All blocks are connected to a single cable, so only one position/length equation will be required. Define position coordinates for each block, write out the position relation, and then differentiate it to relate the velocities.



Group Problem Solving II: Pulley System with 3 Blocks

Solutions:

- 1) A datum line can be drawn through the upper, fixed, pulleys and position coordinates defined from this line to each block (or the pulley above the block).
- 2) Defining s_A , s_B and s_C as shown, the position relation can be written:

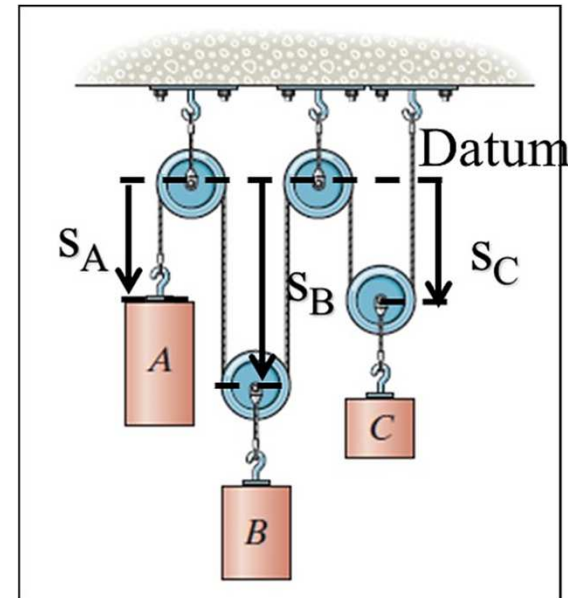
$$s_A + 2s_B + 2s_C = l_T$$

- 3) Differentiate to relate velocities:

$$v_A + 2v_B + 2v_C = 0$$

$$6 + 2v_B + 2(18) = 0$$

$$v_B = -21 \text{ m/s} = 21 \text{ m/s } \uparrow$$



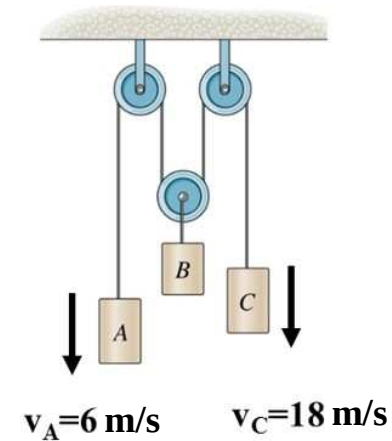
The velocity of block B is 21 m/s up.



Attention Quiz

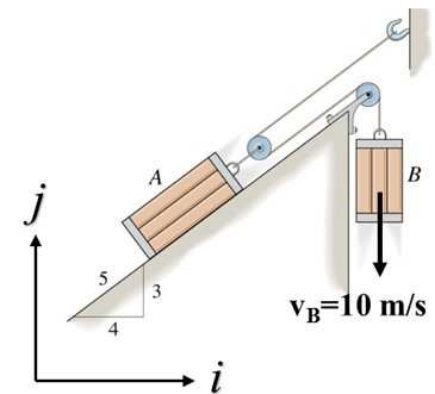
1. Determine the speed of block B when block A is moving down at 6 m/s while block C is moving down at 18 m/s .

A. 24 m/s B. 3 m/s
C. 12 m/s D. 9 m/s

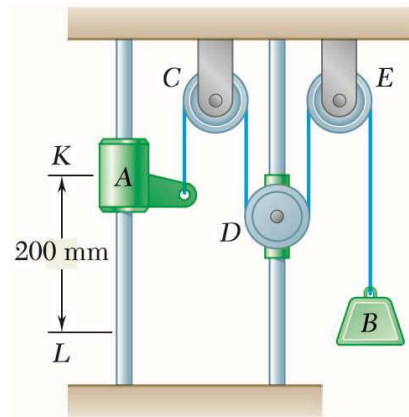


2. Determine the velocity vector of block A when block B is moving downward with a speed of 10 m/s.

A. $(8i + 6j) \text{ m/s}$ B. $(4i + 3j) \text{ m/s}$
C. $(-8i - 6j) \text{ m/s}$ D. $(3i + 4j) \text{ m/s}$



Sample Problem: Pulley and Collar

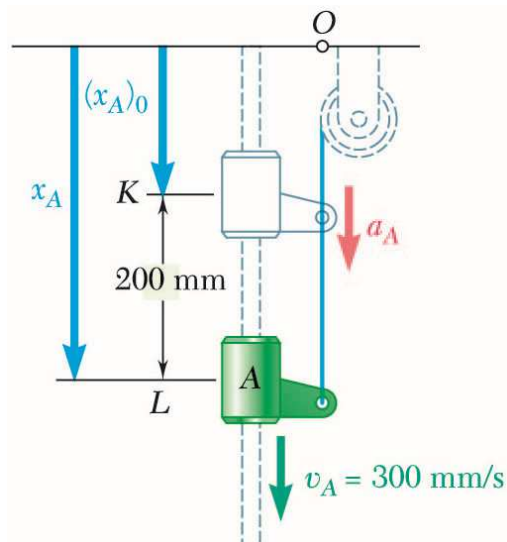


Pulley D is attached to a collar which is pulled down at 75 mm/s . At $t = 0$, collar A starts moving down from K with constant acceleration and zero initial velocity. Knowing that velocity of collar A is 300 mm/s as it passes L , determine the change in elevation, velocity, and acceleration of block B when block A is at L .

STRATEGY:

- Define origin at upper horizontal surface with positive displacement downward.
- Collar A has uniformly accelerated rectilinear motion. Solve for acceleration and time t to reach L .
- Pulley D has uniform rectilinear motion. Calculate change of position at time t .
- Block B motion is dependent on motions of collar A and pulley D . Write motion relationship and solve for change of block B position at time t .
- Differentiate motion relation twice to develop equations for velocity and acceleration of block B .

Sample Problem: Pulley and Collar



MODELING and ANALYSIS:

Define origin at upper horizontal surface with positive displacement downward.

Collar A has uniformly accelerated rectilinear motion. Solve for acceleration and time t to reach L .

$$v_A^2 = (v_A)_0^2 + 2a_A[x_A - (x_A)_0]$$

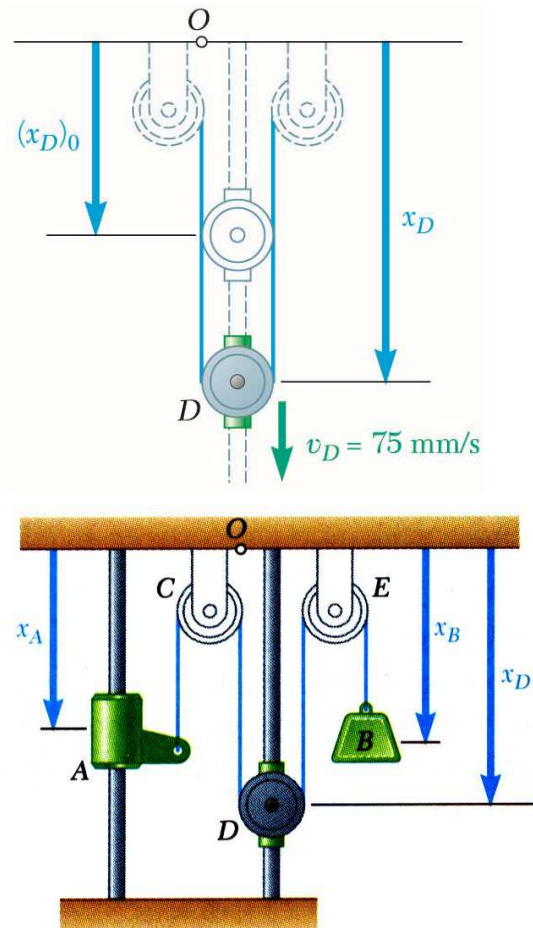
$$\left(300 \frac{\text{mm}}{\text{s}}\right)^2 = 2a_A(200 \text{ mm}) \quad a_A = 225 \frac{\text{mm}}{\text{s}^2}$$

$$v_A = (v_A)_0 + a_A t$$

$$300 \frac{\text{mm}}{\text{s}} = 225 \frac{\text{mm}}{\text{s}^2} t \quad t = 1.333 \text{ s}$$



Sample Problem: Pulley and Collar



- Pulley D has uniform rectilinear motion. Calculate change of position at time t .

$$x_D = (x_D)_0 + v_D t$$

$$x_D - (x_D)_0 = \left(75 \frac{\text{mm}}{\text{s}} \right) (1.333 \text{ s}) = 100 \text{ mm}$$

- Block B motion is dependent on motions of collar A and pulley D . Write motion relationship and solve for change of block B position at time t .

Total length of cable remains constant,

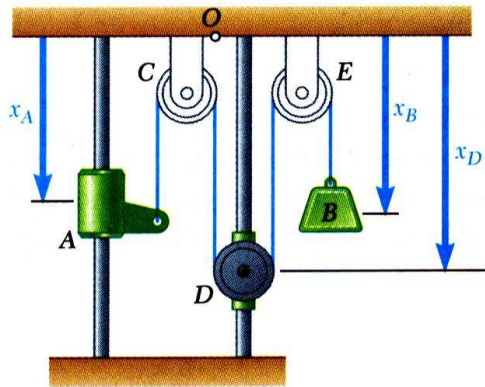
$$x_A + 2x_D + x_B = (x_A)_0 + 2(x_D)_0 + (x_B)_0$$

$$[x_A - (x_A)_0] + 2[x_D - (x_D)_0] + [x_B - (x_B)_0] = 0$$

$$(200 \text{ mm}) + 2(100 \text{ mm}) + [x_B - (x_B)_0] = 0$$

$$x_B - (x_B)_0 = -400 \text{ mm.}$$

Sample Problem: Pulley and Collar



- Differentiate motion relation twice to develop equations for velocity and acceleration of block B .

$$x_A + 2x_D + x_B = \text{constant}$$

$$v_A + 2v_D + v_B = 0$$

$$\left(300 \frac{\text{mm}}{\text{s}}\right) + 2\left(75 \frac{\text{mm}}{\text{s}}\right) + v_B = 0$$

$$v_B = 450 \frac{\text{mm}}{\text{s}}$$

$$a_A + 2a_D + a_B = 0$$

$$\left(225 \frac{\text{mm}}{\text{s}^2}\right) + a_B = 0$$

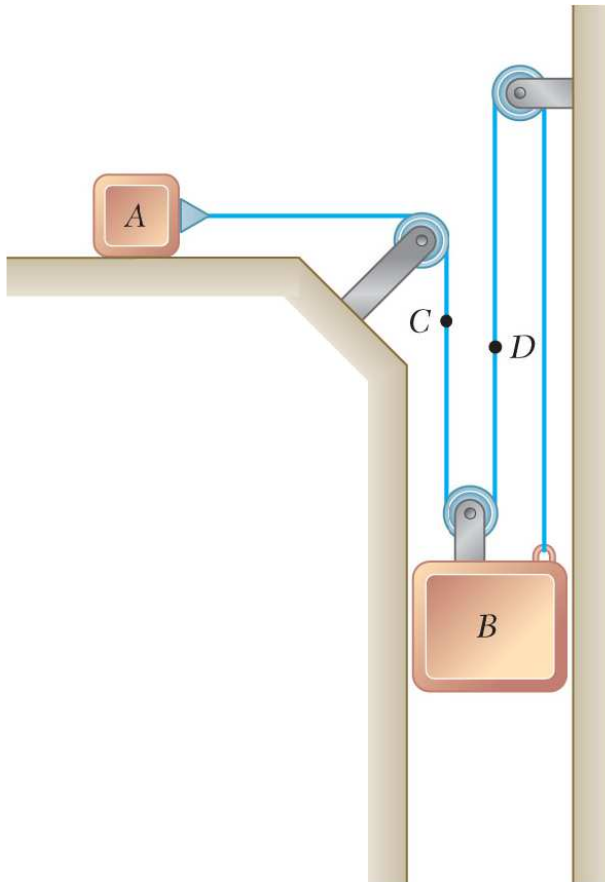
$$a_B = -225 \frac{\text{mm}}{\text{s}^2}$$

REFLECT and THINK:

In this case, the relationship we needed was not between position coordinates, but between changes in position coordinates at two different times. The key step is to clearly define your position vectors. This is a two degree-of-freedom system, because two coordinates are required to completely describe it.



Group Problem Solving: Slider Block and Pulley



Slider block A moves to the left with a constant velocity of 6 m/s . Determine the velocity of block B .

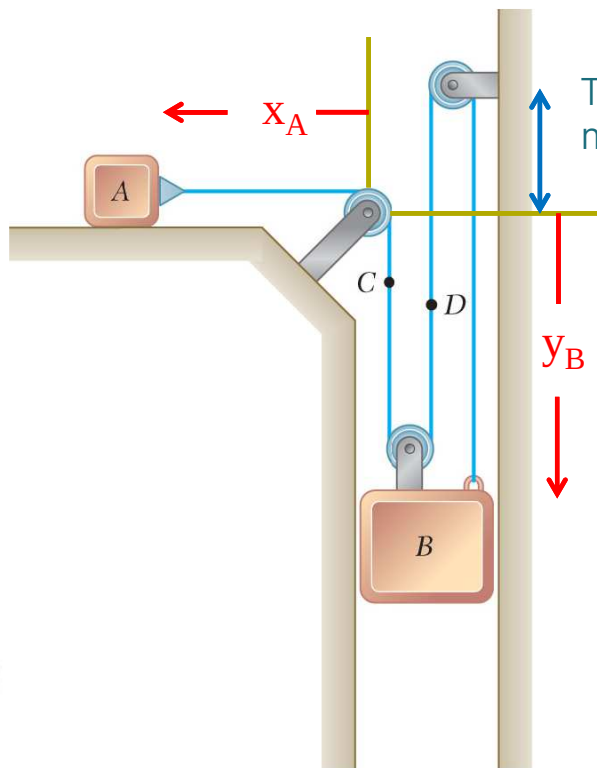
STRATEGY:

- Sketch your system and choose coordinate system
- Write out constraint equation
- Differentiate the constraint equation to get velocity



Group Problem Solving: Slider Block and Pulley

MODELING and ANALYSIS:



This length is constant no matter how the blocks move

Given: $v_A = 6 \text{ m/s}$ left Find: v_B

Sketch your system and choose coordinates

Define your constraint equation(s)

$$x_A + 3y_B + \text{constants} = L$$

Differentiate the constraint equation to get velocity

$$6 \text{ m/s} + 3v_B = 0$$

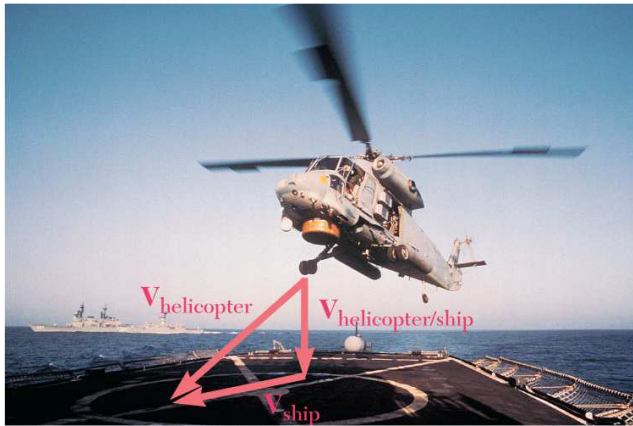
$$v_B = 2 \text{ m/s} \uparrow$$

Note that as x_A gets bigger, y_B gets smaller.

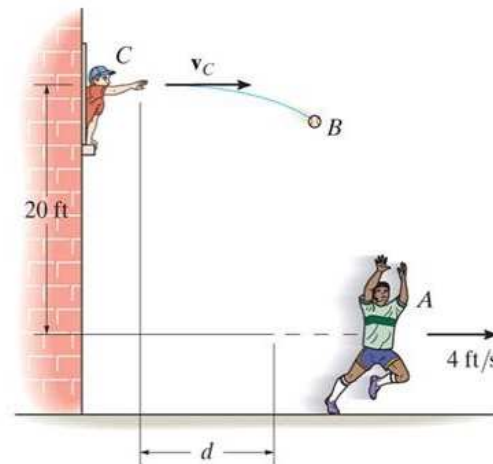


Relative-motion Analysis **Using Translating Axes**

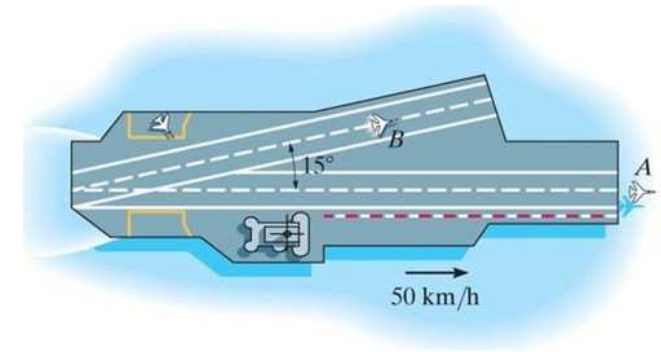
It is critical for a pilot to know the relative motion of his helicopter with respect to the aircraft carrier to make a safe landing.



When you try to hit a moving object, the position, velocity, and acceleration of the object all have to be accounted for by your mind. Here, the boy on the ground is at $d=10$ ft when the girl in the window throws the ball to him. If the boy on the ground is running at a constant speed of 4 feet per second, how fast should the ball be thrown?



When fighter jets take off or land on an aircraft carrier, the velocity of the carrier becomes an issue. If the aircraft carrier is underway with a forward velocity of 50 km/hr and plane A takes off at a horizontal air speed of 200 km/hr (measured by someone on the water), how do we find the velocity of the plane relative to the carrier? How would you find the same thing for airplane B? How does the wind impact this sort of situation?

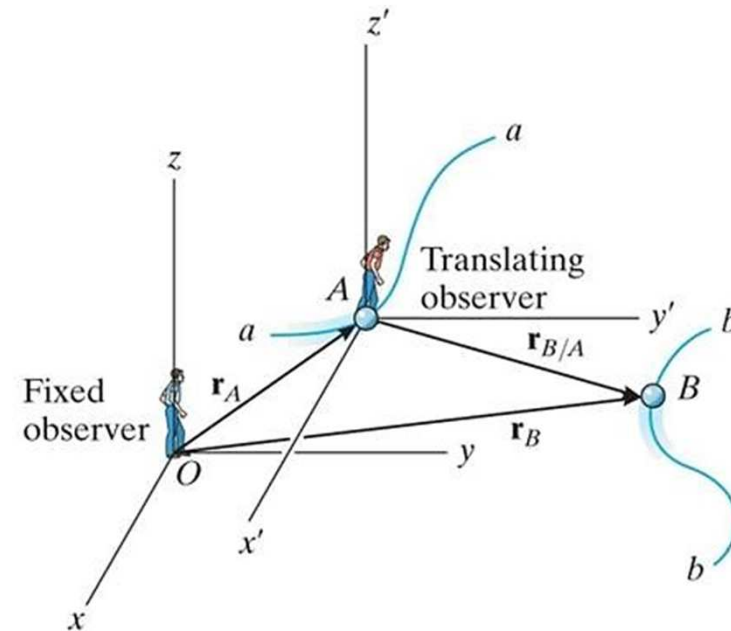


Relative Position

The absolute positions of two particles **A** and **B** with respect to the fixed **x, y, z**-reference frame are given by r_A and r_B .

The position of B relative to A is represented by

$$r_{B/A} = r_B - r_A$$



Therefore, if $r_B = (10i + 2j) m$

and $r_A = (4i + 5j) m$,

then $r_{B/A} = r_B - r_A = (6i - 3j) m$.



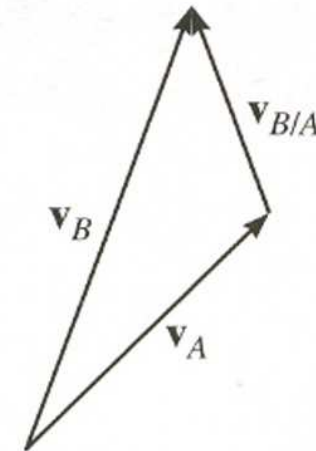
Relative Velocity

To determine the relative velocity of **B** with respect to **A**, the time derivative of the relative position equation is taken.

$$v_{B/A} = v_B - v_A$$

or

$$v_B = v_A + v_{B/A}$$



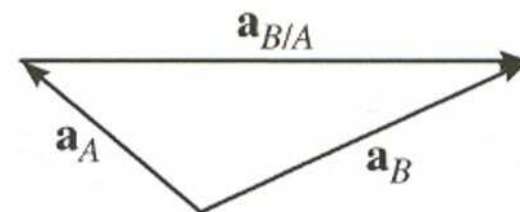
In these equations, v_A and v_B are called absolute velocities and $v_{B/A}$ is the relative velocity of **B** with respect to **A**.

Note that $v_{B/A} = -v_{A/B}$



Relative Acceleration

The time derivative of the relative velocity equation yields a similar vector relationship between the absolute and relative accelerations of particles A and B.



These derivatives yield: $\mathbf{a}_{B/A} = \mathbf{a}_B - \mathbf{a}_A$
or
 $\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$



Solving Problems

Since the relative motion equations are vector equations, problems involving them may be solved in one of two ways.

For instance, the velocity vectors in $v_B = v_A + v_{B/A}$ could be written as two dimensional (2-D) Cartesian vectors and the resulting 2-D scalar component equations solved for up to two unknowns.

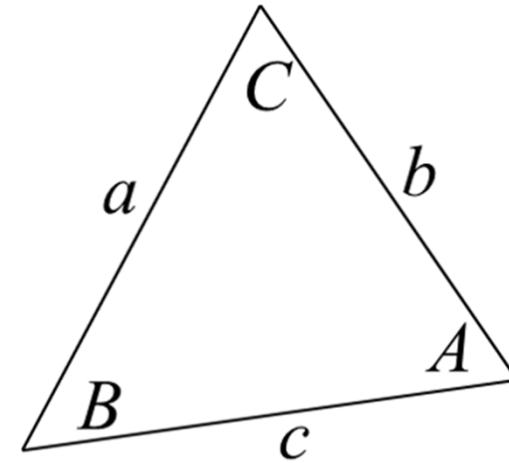
Alternatively, vector problems can be solved “graphically” by use of trigonometry. This approach usually makes use of the law of sines or the law of cosines.

Could a CAD system be used to solve these types of problems?



Laws of Sines and Cosines

Since vector addition or subtraction forms a triangle, sine and cosine laws can be applied to solve for relative or absolute velocities and accelerations. As a review, their formulations are provided below.



Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

Law of Cosines:

$$\begin{aligned}a^2 &= b^2 + c^2 - 2bc \cos(A) \\b^2 &= a^2 + c^2 - 2ac \cos(B) \\c^2 &= a^2 + b^2 - 2ab \cos(C)\end{aligned}$$



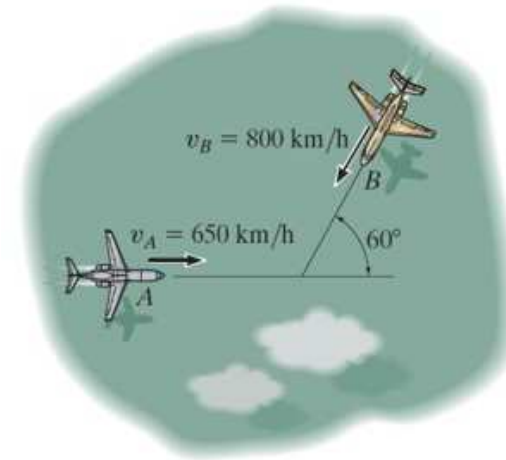
Example: Two Aircraft

Given: Two aircraft as shown.

$$\begin{aligned}v_A &= 650 \text{ km/h} \\v_B &= 800 \text{ km/h}\end{aligned}$$

Find: $v_{B/A}$

- Plan:**
- 1) Vector Method: Write vectors v_A and v_B in Cartesian form, then determine $v_B - v_A$
 - 2) Graphical Method: Draw vectors v_A and v_B from a common point. Apply the laws of sines and cosines to determine $v_{B/A}$.



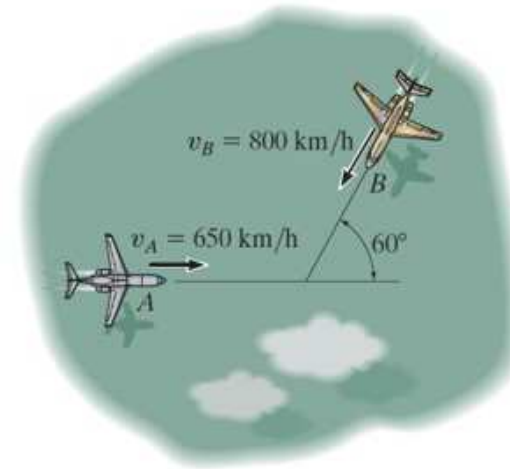
Example: Two Aircraft

Solution:

1) Vector Method

$$v_A = (650 i) \text{ km/h}$$

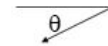
$$\begin{aligned} v_B &= -800 \cos 60 i - 800 \sin 60 j \\ &= (-400 i - 692.8 j) \text{ km/h} \end{aligned}$$



$$v_{B/A} = v_B - v_A = (-1050 i - 692.8 j) \text{ km/h}$$

$$v_{B/A} = \sqrt{(-1050)^2 + (-692.8)^2} = 1258 \text{ km/h}$$

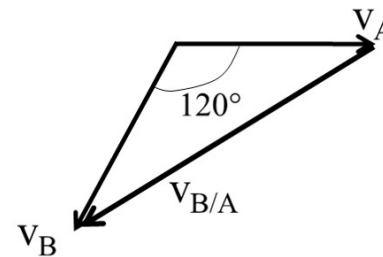
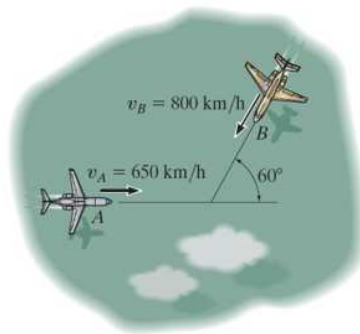
$$\theta = \tan^{-1} \left(\frac{692.8}{1050} \right) = 33.4^\circ$$



Example: Two Aircraft

2) Graphical Method:

Note that the vector that measures the tip of B relative to A is



Law of Cosines: $(v_{B/A})^2 = (800)^2 + (650)^2 - (800)(650) \cos 120^\circ$
 $v_{B/A} = 1258 \text{ km/h}$

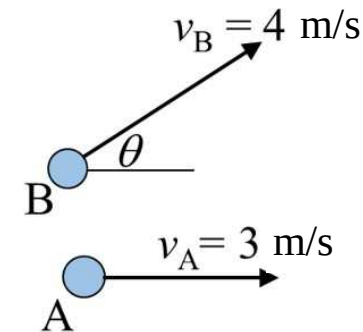
Law of Sines: $\frac{v_{B/A}}{\sin(120^\circ)} = \frac{v_A}{\sin \theta}$ or $\theta = 33.4^\circ$



Concept Quiz

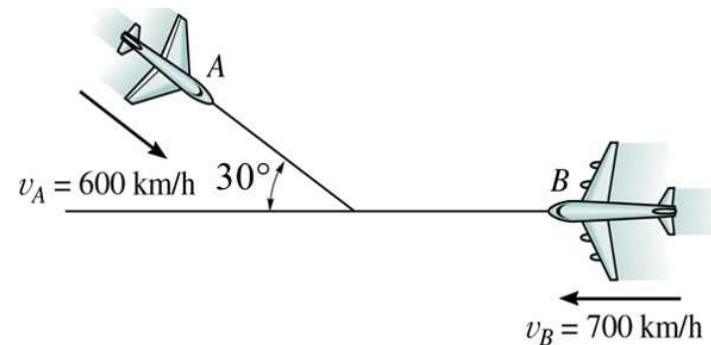
- Two particles, A and B, are moving in the directions shown. What should be the angle θ so that $v_{B/A}$ is minimum?

- A. 0° B. 180°
C. 90° D. 270°



- Determine the velocity of plane A with respect to plane B.

- A. $(400i + 520j) \text{ km/hr}$
B. $(1200i - 300j) \text{ km/hr}$
C. $(-181i - 300j) \text{ km/hr}$
D. $(-1200i + 300j) \text{ km/hr}$



Group Problem Solving: Expressway Exit

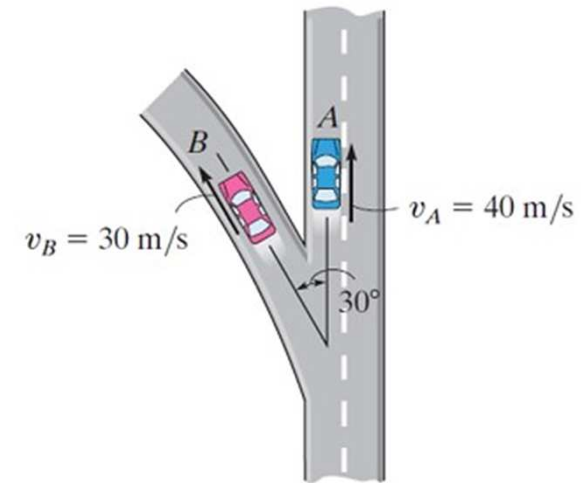
Given: Car A moves in a straight line while Car B moves along a curve having a radius of curvature of 200 meter.

$$v_A = 40 \text{ m/s}^2$$

$$v_B = 30 \text{ m/s}^2$$

$$a_A = 4 \text{ m/s}^2$$

$$a_B = -3 \text{ m/s}^2$$



Find: $v_{B/A}$ $a_{B/A}$

Plan: Write the velocity and acceleration vectors for Cars A and B. Determine $v_{B/A}$ and $a_{B/A}$ by using vector relationships.



Group Problem Solving: Expressway Exit

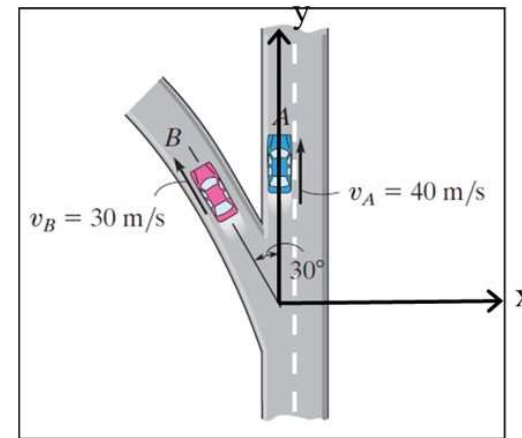
Solution:

The velocity of B is:

$$\begin{aligned} v_B &= -30 \sin(30^\circ)i + 30 \cos(30^\circ)j \\ &= (-15i + 25.98j) \text{ m/s} \end{aligned}$$

The velocity of A is:

$$v_A = 40j \text{ (m/s)}$$



The relative velocity of B with respect to A is $(v_{B/A})$.

$$v_{B/A} = v_B - v_A = (-15i + 25.98j) - (40j) = (-15i - 14.02j) \text{ m/s}$$

$$\text{or } v_{B/A} = \sqrt{(-15)^2 + (-14.02)^2} = 20.5 \text{ m/s}$$

$$\theta = \tan^{-1}\left(\frac{14.02}{15}\right) = 43.1^\circ$$



Group Problem Solving: Expressway Exit

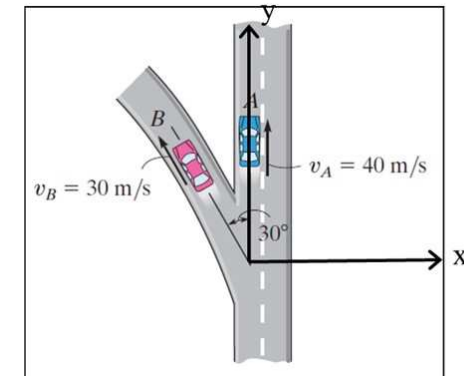
Since car B is traveling along a curve, the acceleration of B is:

$$\begin{aligned} \mathbf{a}_B &= (\mathbf{a}_t)_B + (\mathbf{a}_n)_B = [-(-3) \sin(30^\circ) \mathbf{i} + (-3) \cos(30^\circ) \mathbf{j}] \\ &\quad + \left[- \left(\frac{30^2}{200} \right) \cos(30^\circ) \mathbf{i} - \left(\frac{30^2}{200} \right) \sin(30^\circ) \mathbf{j} \right] \\ \mathbf{a}_B &= (-2.397 \mathbf{i} - 4.848 \mathbf{j}) \text{ m/s}^2 \end{aligned}$$

The acceleration of A is: $\mathbf{a}_A = (4 \mathbf{j}) \text{ m/s}^2$

The relative acceleration of B with respect to A is:

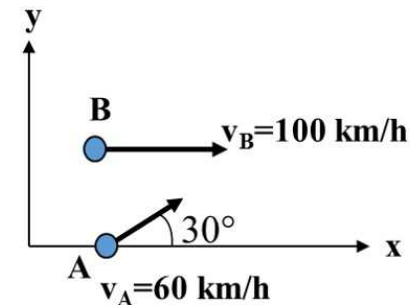
$$\begin{aligned} \mathbf{a}_{B/A} &= \mathbf{a}_B - \mathbf{a}_A = (-2.397 \mathbf{i} - 4.848 \mathbf{j}) \text{ m/s}^2 \\ a_{B/A} &= \sqrt{(-2.397)^2 + (-8.848)^2} \\ &= 9.17 \text{ m/s}^2 \\ \beta &= \tan^{-1}(8.848/2.397) = 74.8^\circ \end{aligned}$$



Attention Quiz

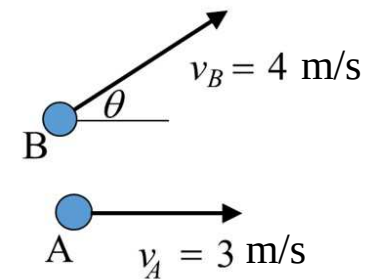
- Determine the relative velocity of particle B with respect to particle A.

- $(48i + 30j)$ km/hr
- $(-40i + 30j)$ km/hr
- $(48i - 30j)$ km/hr
- $(-40i - 30j)$ km/hr

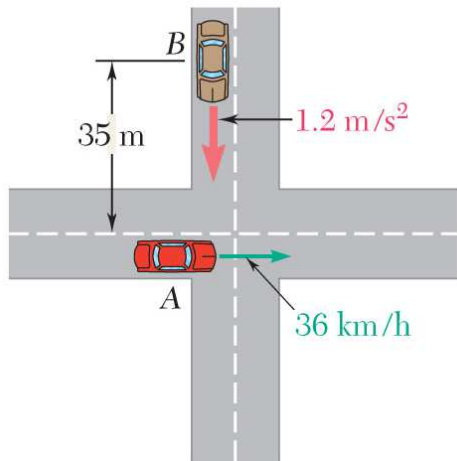


- If theta equals 90° and A and B start moving from the same point, what is the magnitude of $r_{B/A}$ at $t = 5$ s?

- 20 meter
- 15 meter
- 18 meter
- 25 meter



Sample Problem: Cars near Intersection



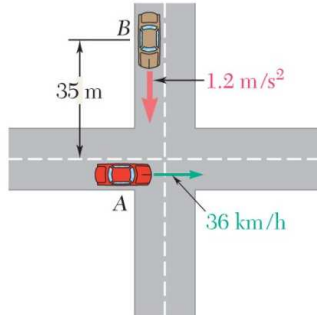
Automobile *A* is traveling east at the constant speed of 36 km/h. As automobile *A* crosses the intersection shown, automobile *B* starts from rest 35 m north of the intersection and moves south with a constant acceleration of 1.2 m/s^2 . Determine the position, velocity, and acceleration of *B* relative to *A* 5 s after *A* crosses the intersection.

STRATEGY:

- Define inertial axes for the system
- Determine the position, speed, and acceleration of car *A* at $t = 5 \text{ s}$
- Determine the position, speed, and acceleration of car *B* at $t = 5 \text{ s}$
- Using vectors or a graphical approach, determine the relative position, velocity, and acceleration



Sample Problem: Cars near Intersection



MODELING and ANALYSIS:

- Define axes along the road

Given: $v_A = 36 \text{ km/h}$, $a_A = 0$, $(x_A)_0 = 0$

$(v_B)_0 = 0$, $a_B = -1.2 \text{ m/s}^2$, $(y_A)_0 = 35 \text{ m}$

Determine motion of Automobile A:

$$v_A = \left(36 \frac{\text{km}}{\text{h}} \right) \left(\frac{1000 \text{ m}}{1 \text{ km}} \right) \left(\frac{1 \text{ h}}{3600 \text{ s}} \right) = 10 \text{ m/s}$$

We have uniform motion for A so:

$$a_A = 0$$

$$v_A = +10 \text{ m/s}$$

$$x_A = (x_A)_0 + v_A t = 0 + 10t$$

At $t = 5 \text{ s}$

$$a_A = 0$$

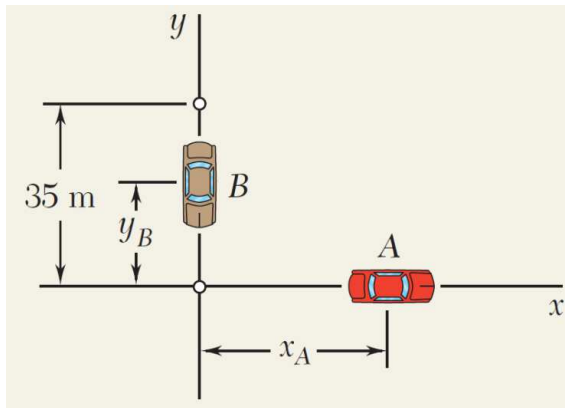
$$v_A = +10 \text{ m/s}$$

$$x_A = +(10 \text{ m/s})(5 \text{ s}) = +50 \text{ m}$$

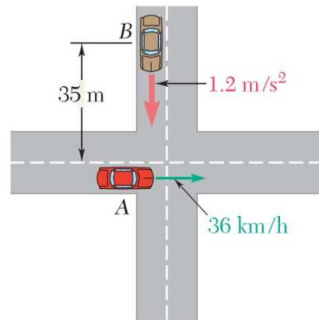
$$\mathbf{a}_A = 0$$

$$\mathbf{v}_A = 10 \text{ m/s} \rightarrow$$

$$\mathbf{r}_A = 50 \text{ m} \rightarrow$$



Sample Problem: Cars near Intersection



MODELING and ANALYSIS:

Determine motion of Automobile B:

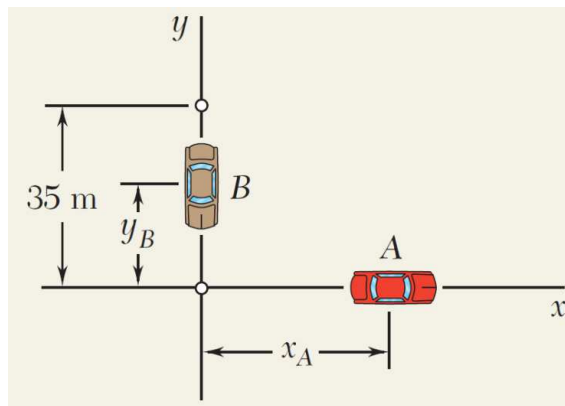
We have uniform acceleration for B so:

$$\begin{aligned} a_B &= -1.2 \text{ m/s}^2 \\ v_B &= (v_B)_0 + at = 0 - 1.2 t \\ y_B &= (y_B)_0 + (v_B)_0 t + \frac{1}{2} a_B t^2 = 35 + 0 - \frac{1}{2}(1.2)t^2 \end{aligned}$$

At $t = 5 \text{ s}$

$$\begin{aligned} a_B &= -1.2 \text{ m/s}^2 \\ v_B &= -(1.2 \text{ m/s}^2)(5 \text{ s}) = -6 \text{ m/s} \\ y_B &= 35 - \frac{1}{2}(1.2 \text{ m/s}^2)(5 \text{ s})^2 = +20 \text{ m} \end{aligned}$$

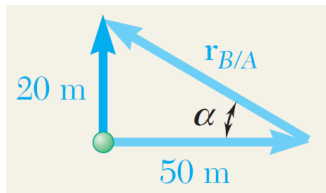
$$\begin{aligned} \mathbf{a}_B &= 1.2 \text{ m/s}^2 \downarrow \\ \mathbf{v}_B &= 6 \text{ m/s} \downarrow \\ \mathbf{r}_B &= 20 \text{ m} \uparrow \end{aligned}$$



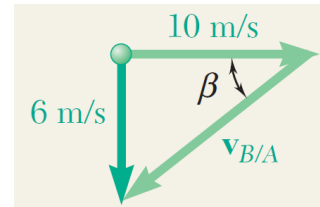
Sample Problem: Cars near Intersection

$$\begin{aligned} \mathbf{a}_A &= 0 & \mathbf{a}_B &= 1.2 \text{ m/s}^2 \downarrow \\ \mathbf{v}_A &= 10 \text{ m/s} \rightarrow & \mathbf{v}_B &= 6 \text{ m/s} \downarrow \\ \mathbf{r}_A &= 50 \text{ m} \rightarrow & \mathbf{r}_B &= 20 \text{ m} \uparrow \end{aligned}$$

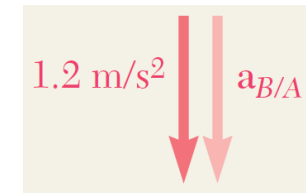
We can solve the problems geometrically, and apply the arctangent relationship:



$$\begin{aligned} r_{B/A} &= 53.9 \text{ m} & \alpha &= 21.8^\circ \\ \mathbf{r}_{B/A} &= 53.9 \text{ m} \nearrow 21.8^\circ \end{aligned}$$



$$\begin{aligned} v_{B/A} &= 11.66 \text{ m/s} & \beta &= 31.0^\circ \\ \mathbf{v}_{B/A} &= 11.66 \text{ m/s} \nearrow 31.0^\circ \end{aligned}$$



$$\mathbf{a}_{B/A} = 1.2 \text{ m/s}^2 \downarrow$$

Or we can solve the problems using vectors to obtain equivalent results:

$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A}$$

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$$

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$$

$$20\mathbf{j} = 50\mathbf{i} + \mathbf{r}_{B/A}$$

$$-6\mathbf{j} = 10\mathbf{i} + \mathbf{v}_{B/A}$$

$$-1.2\mathbf{j} = 0\mathbf{i} + \mathbf{a}_{B/A}$$

$$\mathbf{r}_{B/A} = 20\mathbf{j} - 50\mathbf{i} \text{ (m)}$$

$$\mathbf{v}_{B/A} = -6\mathbf{j} - 10\mathbf{i} \text{ (m/s)}$$

$$\mathbf{a}_{B/A} = -1.2\mathbf{j} \text{ (m/s}^2\text{)}$$

Physically, a rider in car A would "see" car B travelling south and west.



Concept Quiz

If you are sitting in train B looking out the window, in which direction does it appear that train A is moving?

